MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, OPERATIONAL, ANALOG, KEY CODE FGC

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specification and standards. Unless otherwise specified, the following specification and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of the specification and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.385 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FGC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/177-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 15.0 ±1.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 16.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

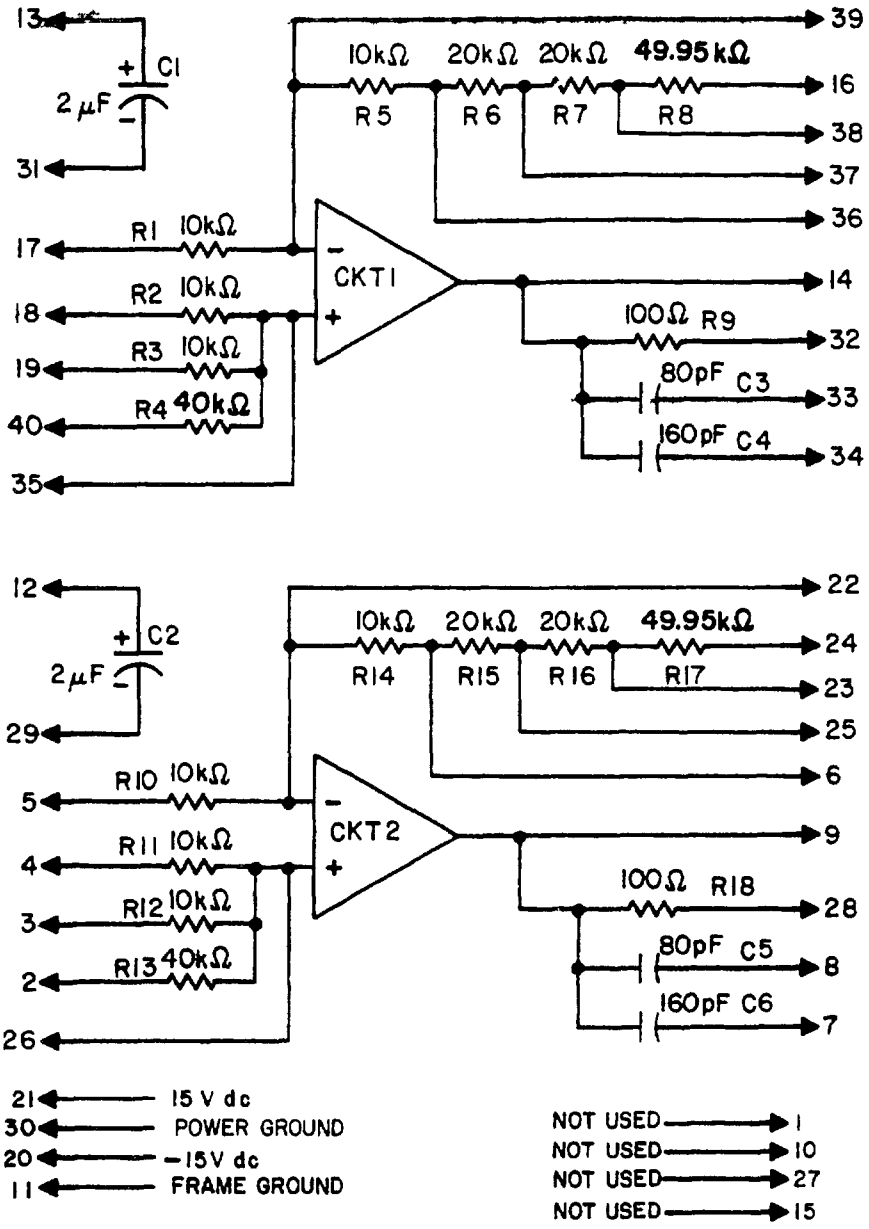
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. All resistors are ±0.3%, 1/8 W, except the 100Ω is ±10%, 1/4 W, and the 49.95 kΩ is ±0.4%, 1/8 W.
2. All capacitors are ±5%, 18 V

* FIGURE 1. FGC function diagram, operational amplifier.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	V_S stress	III	Q	-80	80	-80	80	-80	80	-80	80	mV
B1-B4	Output short circuit current	IV	S	1	40	1	45	1	45	1	45	mA
C1-C7	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1, D2	Power supply current	VI	S	0.5	8.0	0.5	8.0	0.5	8.0	0.5	8.0	mA
E1-E4	Input offset voltage	VII	X	-5	5	-6	6	-6	6	-6	6	mV
F1, F2	Input offset current	VII	X	-50	50	-50	50	-50	50	-50	50	nA
G1, G2	Large signal voltage gain	VIII	X	50	290	50	290	50	290	50	290	V/mV
H1, H2	Gain functional	IX	X	-10.08	-9.92	-10.09	-9.91	-10.09	-9.91	-10.09	-9.91	V
J1-J4	Slew rate	X	X	1.2	100	1.2	100	1.2	100	1.2	100	V/µs
K1, K2	Common mode rejection ratio	XI	X	-∞	-80	-∞	-80	-∞	-80	-∞	-80	dB
L1, L2	Supply voltage rejection ratio	XII	S	-∞	-66	-∞	-66	-∞	-66	-∞	-66	dB
M1, M2	Small signal bandwidth	XIII	S	3	10	3	10	3	10	3	10	MHz
M3, M4	Gain peaking ratio	XIII	S	1.0	2.5	1.0	2.5	1.0	2.5	1.0	2.5	V/V
N1, N2	Output voltage swing, positive	XIV	S	8.5	13.5	8.5	13.5	8.5	13.5	8.5	13.5	Vpk
N3, N4	Output voltage swing, negative	XIV	S	-13.5	-8.5	-13.5	-8.5	-13.5	-8.5	-13.5	-8.5	Vpk
P1, P2	Noise	XV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mV rms
R1, R2	Capacitance (C1 and C2)	XVI	X	1.9	2.1	1.9	2.1	1.9	2.1	1.9	2.1	µF
R3, R4	Capacitance (C3 and C5)	XVI	X	76	84	76	84	76	84	76	84	pF

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R5, R6	Capacitance (C4 and C6)	XVI	X	152	168	152	168	152	168	152	168	pF
R7, R8	Dissipation factor (C1 and C2)	XVI	X	0	5	0	5	0	5	0	5	%
R9-R12	Dissipation factor (C3-C6)	XVI	X	0	3	0	3	0	3	0	3	%
R13, R14	Power supply capacitance	XVI	X	0.075	0.125	0.075	0.125	0.075	0.125	0.075	0.125	µF
S1, S6, S7, S13, S18, S19	Resistance (R1, R2, R3, R10, R11, and R12)	XVII	X	9.98	10.02	9.97	10.03	9.97	10.03	9.97	10.03	kΩ
S2, S11, S14, S23	Resistance (R1+R5, R2+R3, R10+R14, and R11+R12)	XVII	X	19.96	20.04	19.94	20.06	19.94	20.06	19.94	20.06	kΩ
S3, S8, S15, S20	Resistance (R1+R5+R6, R4, R10+R14+R15, and R13)	XVII	X	39.92	40.08	39.88	40.12	39.88	40.12	39.88	40.12	kΩ
S4, S16	Resistance (R1+R5+R6+R7 and R10+R14+R15+R16)	XVII	X	59.88	60.12	59.82	60.18	59.82	60.18	59.82	60.18	kΩ
S5, S17	Resistance (R1+R5+R6+R7+R8 and R10+R14+R15+R16+R17)	XVII	X	109.68	110.22	109.57	110.33	109.57	110.33	109.57	110.33	kΩ
S9, S10, S21, S22	Resistance (R3+R4, R2+R4, R12+R13, and R11+R13)	XVII	X	49.90	50.10	49.85	50.15	49.85	50.15	49.85	50.15	kΩ
S12, S24	Resistance (R9 and R18)	XVII	X	95.0	105.0	90.0	110.0	90.0	110.0	90.0	110.0	Ω
T	Frame continuity	XVIII	S	0	1	0	1	0	1	0	1	Ω
U1, U2	Average input bias current (each input)	XIX	X	0	120	0	120	0	120	0	120	nA
V1, V2	Output latch	Fig 4	Q	---	---	-66	66	---	---	-66	66	mV

1/ (X) parameters shall be 100 percent tested during module acceptance tests.
2/ (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ (Q) parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test, and prior to disconnecting the power, electrical tests V1 and V2 shall be performed in accordance with figure 4. While still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/177-1 supersedes part number M28787/177. The two parts are interchangeable.

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits of table I.

Supply voltage	±15 V dc ±10%
Maximum power dissipation	
+16.5 V supply	132 mW (no signal)
-16.5 V supply	132 mW (no signal)
Input offset voltage	6 mV maximum
Input offset current	50 nA maximum
Input bias current	120 nA (average, each input)
Common mode rejection ratio	-80 dB maximum
Supply voltage rejection ratio	-66 dB maximum
Slew rate (unity gain)	1.2 V/µs minimum ($R_L \geq$ 2kΩ, V_S = ±15 V)
Small signal bandwidth (unity gain)	3 MHz minimum
Output voltage swing	
V_S = ±13.5 V	±8.5 V peak (R_L = 569Ω, f = 20 kHz)
Output-drive current capability	
V_O = 8.5 V peak	15 mA minimum
Large signal voltage gain	50,000 minimum
Gain options	-3.5 dB minimum to 20.0 dB maximum
Gain accuracy	±1 percent ($f_O \leq$ 20 kHz, $R_L \geq$ 2 kΩ)
Noise	1 mV rms maximum (input at ground, output open, gain = 10)

* 6.5 Module function. This module contains two independent general purpose amplifiers and two independent 2 µF capacitors.

*6.5.1 Input to output relationship. Each amplifier circuit includes, among others, the gain options specified in table II.

TABLE II. FGC module gain options 1/

R_f	Inverting		Noninverting	
kΩ	G	G(dB)	G	G(dB)
10	1.0	0	1.0	0
20	2.0	6.02	1.5	3.52
30	3.0	9.54	2.0	6.02
40	4.0	12.04	2.5	7.95
50	5.0	14	3.0	9.55
60	6.0	15.6	3.5	10.9
70	7.0	16.9	4.0	12.04
80	8.0	18.06	4.5	13.05
90	9.0	19.1	5.0	13.95
100	10.0	20.0	5.5	14.8
6.67	0.667	-3.5	0.834	-1.6
12.0	1.2	1.58	1.1	0.827
21.0	2.1	6.45	1.55	3.8
25.0	2.5	7.96	1.75	4.86
25.6	2.56	8.16	1.78	5.01
32.2	3.22	10.15	2.11	6.48
44.3	4.43	12.9	2.72	8.7

1/ By choosing different input configurations, the inverting gains (G) may be scaled by 2.0, 1.6, 0.4, or 1.78.

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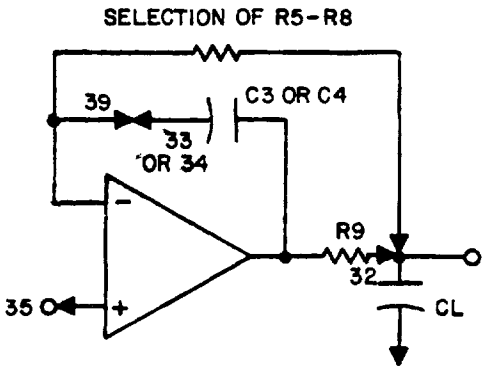
* 6.5.2 Stability. A capacitive load in conjunction with the open loop output resistance of the amplifier may introduce another pole in the gain function at $S = \frac{1}{R_O C_L}$ (where R_O is the operational amplifier's open loop output impedance and C_L is the capacitive load external to the amplifier). This may cause an additional "break" in the frequency response and a rolloff of -12 dB per octave above the frequency $W = \frac{1}{R_O C_L}$. If the closed loop gain curve intersects this section of the effective open-loop gain curve, the amplifier may be marginally stable. At low gain levels, the additional pole can cause peaking, ringing, or even oscillation unless further phase compensation correction is made. Since the internal phase compensation is not accessible for modification, it may be added in the feedback loop as shown on figure 2 using components available in the module (the designation used are for circuit 1 of figure 1).

* 6.5.3 Power supply voltage. The module will meet the requirements of table I for a power supply voltage (V_S) of ±15.0±1.5 V dc when driving the specified loads.

* 6.5.4 Parallel outputs. Paralleling of outputs is not permissible.

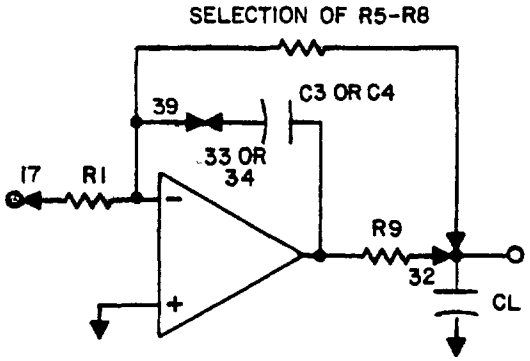
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Non-inverting

R9, R5-R8, and C3 or C4 connected for compensation

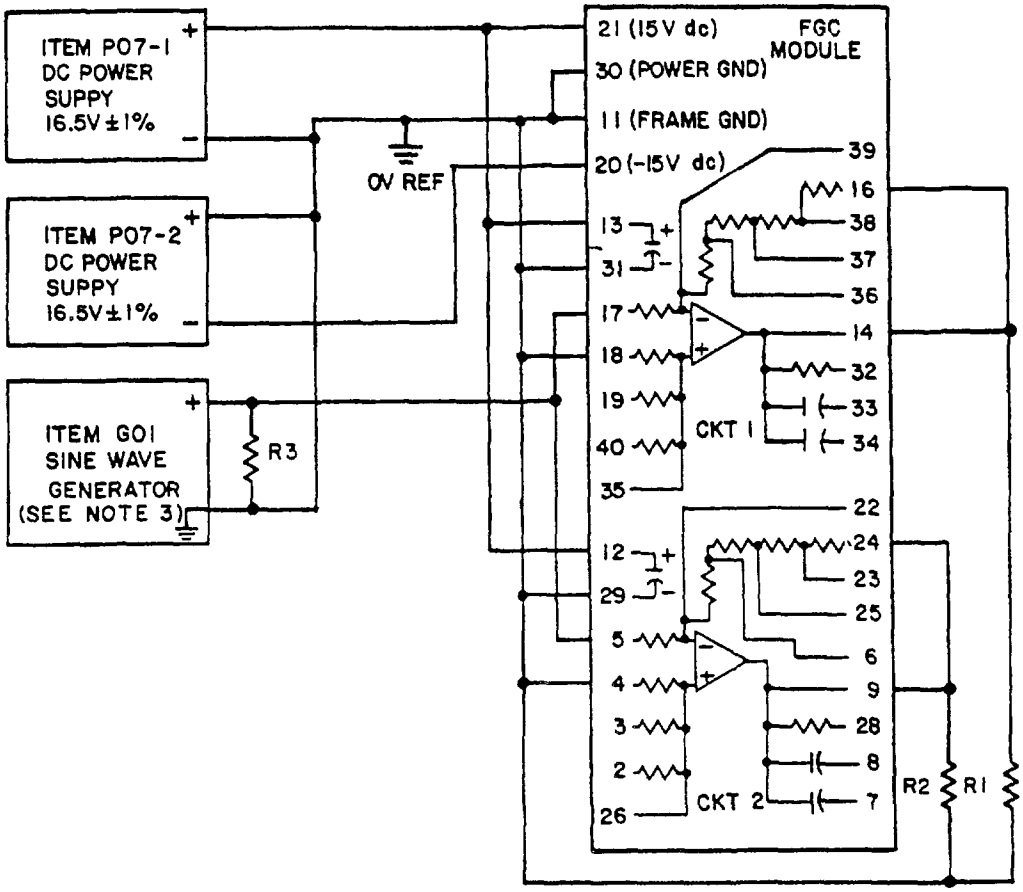


Inverting

R9 and C3 or C4 connected for compensation. R5-R8 and R1 determine the users required gain.

* FIGURE 2. Stability compensation diagrams

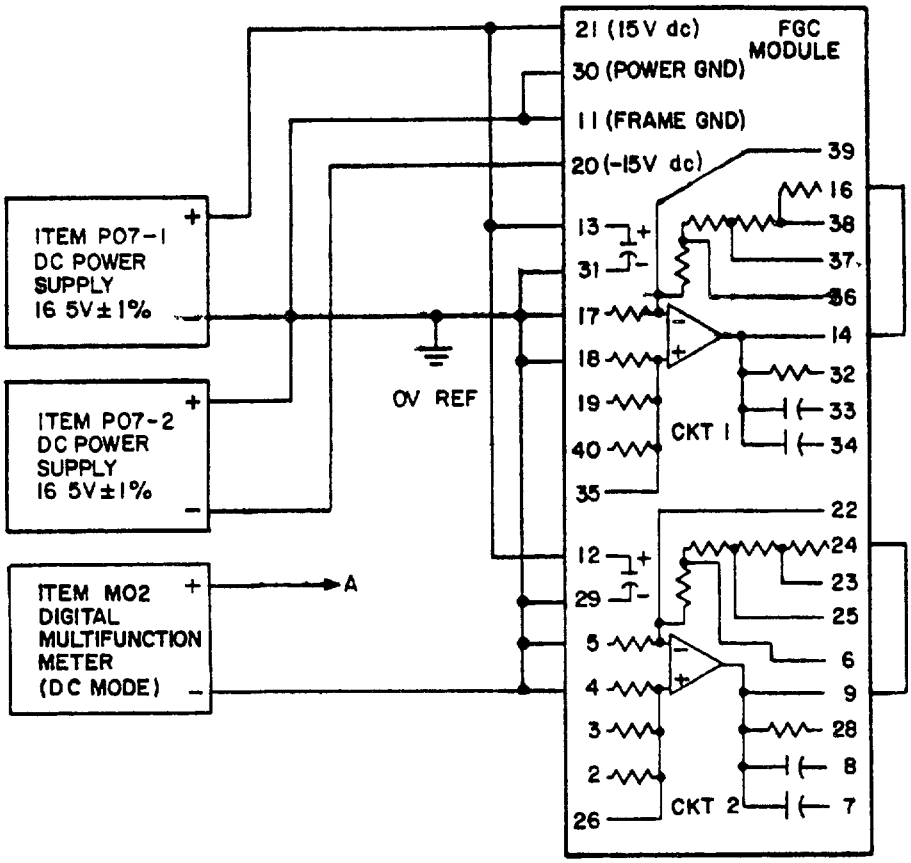
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NOTES:
1. R1 = R2 = 569Ω ±1%, 1/4 W.
2. R3 = 49.9Ω ±5%, 1 W.
3. Item G01 setup
   Amplitude = 2.0 ±0.1 V p-p.
   Frequency = 1 kHz ±2%.

* FIGURE 3. Operating temperature and component temperature test circuit

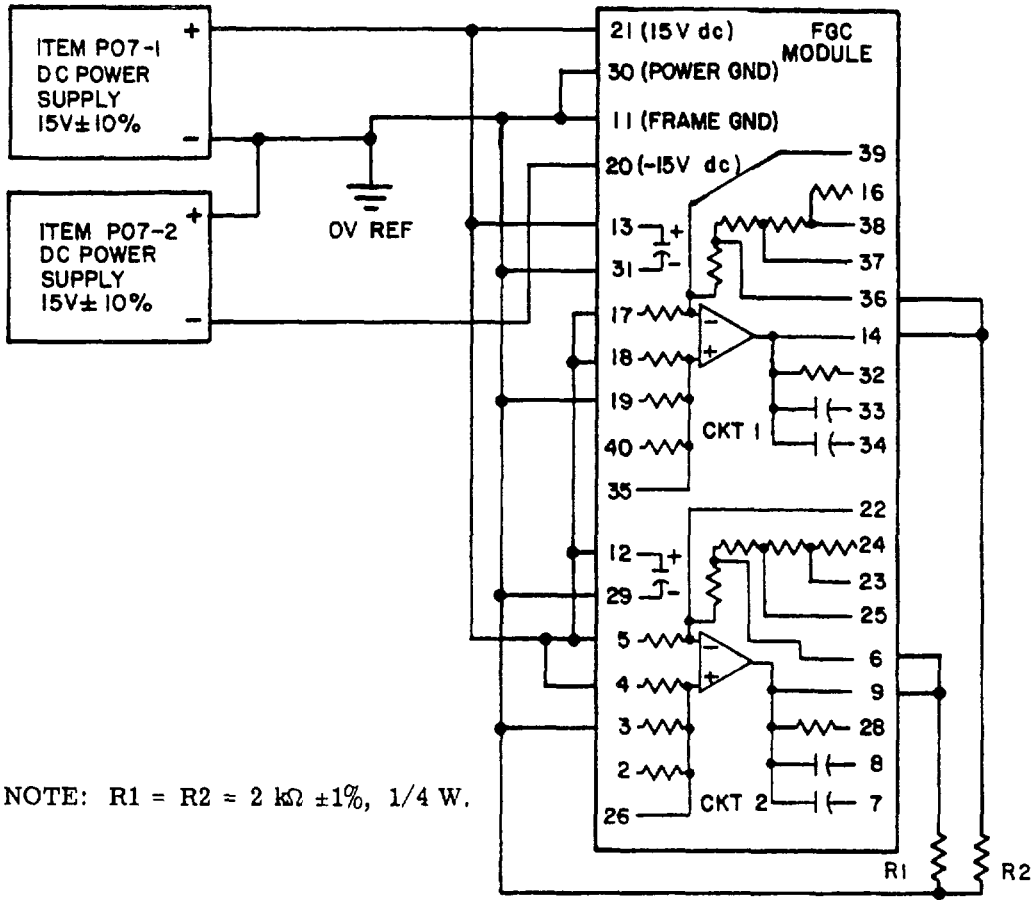
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NOTES.
1. Following the life test period, prior to disconnecting the power, connect point A to pin 14. The voltage shall meet the limits specified in table I for test condition VI.
2. For test condition V2, connect point A to pin 9.
3 Test condition V verifies that the amplifiers do not latch up.

FIGURE 4. Life test circuit.

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NOTE: R1 = R2 = 2 kΩ ±1%, 1/4 W.

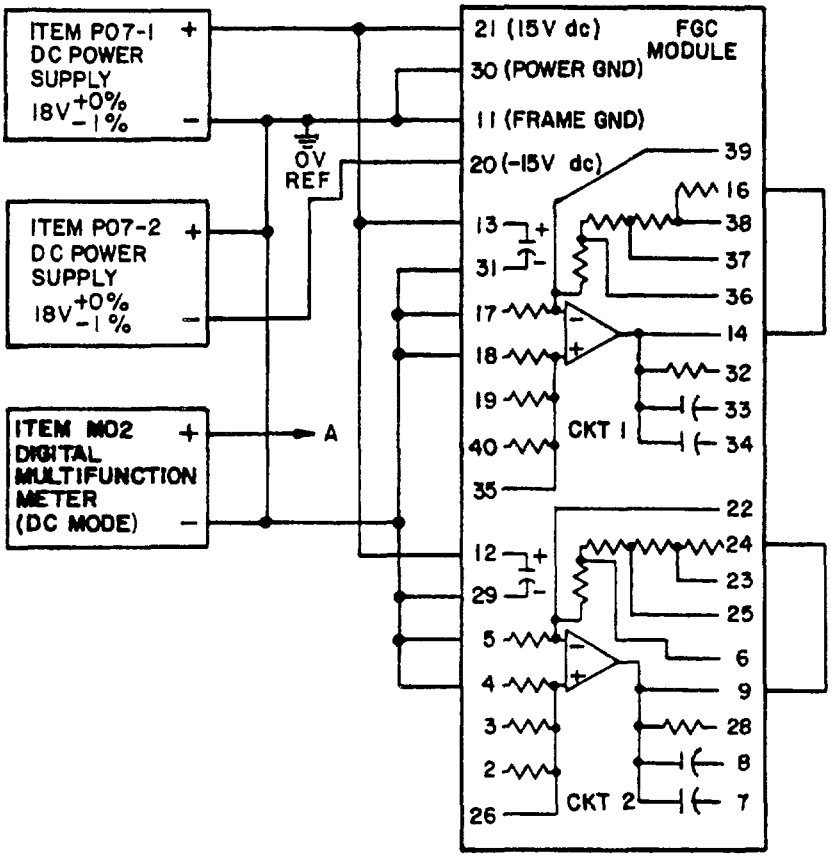
FIGURE 5. Vibration test circuit.

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TABLE III. V_S stress test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	V_S stress	14	-80	80	mV	6
A2	V_S stress	9	-80	80	mV	6

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NOTE: Measurement shall be recorded after point A has been connected to the appropriate module pin for a minimum of 1 minute.

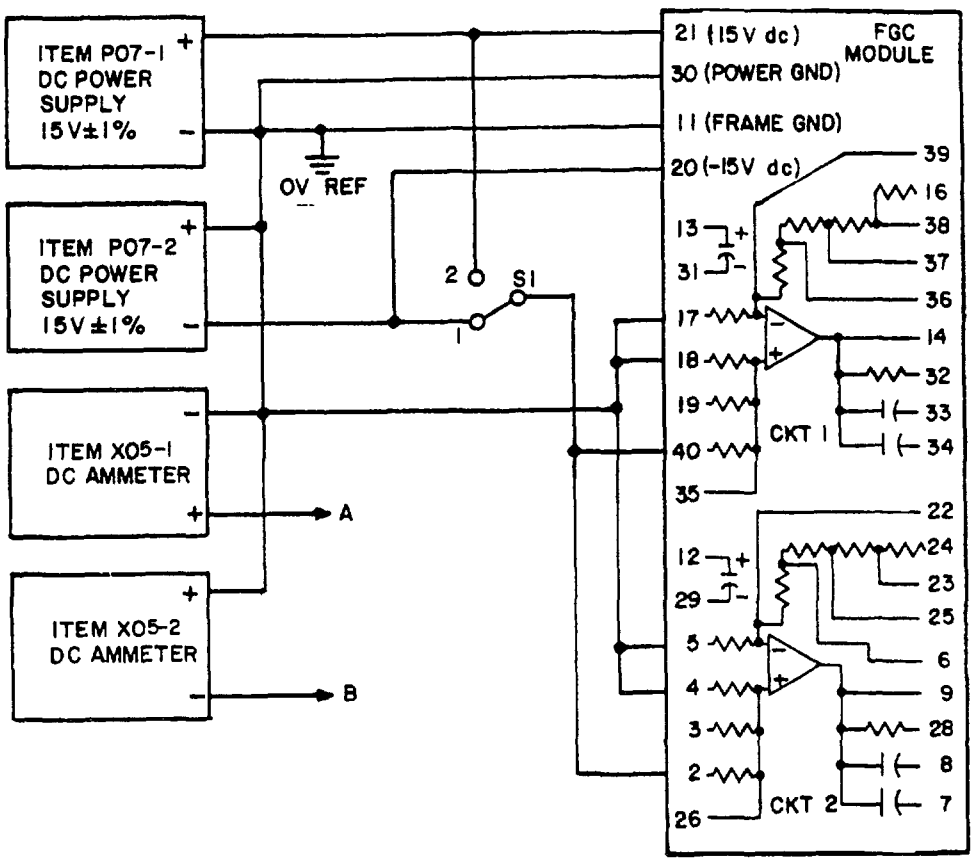
FIGURE 6. V_S stress test circuit for table III

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TABLE IV. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	S1 position	Min	Max		
B1	Output short circuit current (-IN)	Open	16, 14	1	1	40	mA	7
B2	Output short circuit current (+IN)	16, 14	Open	2	"	"	"	"
B3	Output short circuit current (-IN)	Open	24, 9	1	"	"	"	"
B4	Output short circuit current (+IN)	24, 9	Open	2	"	"	"	"

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NOTES:
1. S1 = SPDT switch.
2. With points A and B disconnected, item X05 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken after 1 minute
4. Since items X05-1 and X05-2 are not used simultaneously, only one item X05 is required

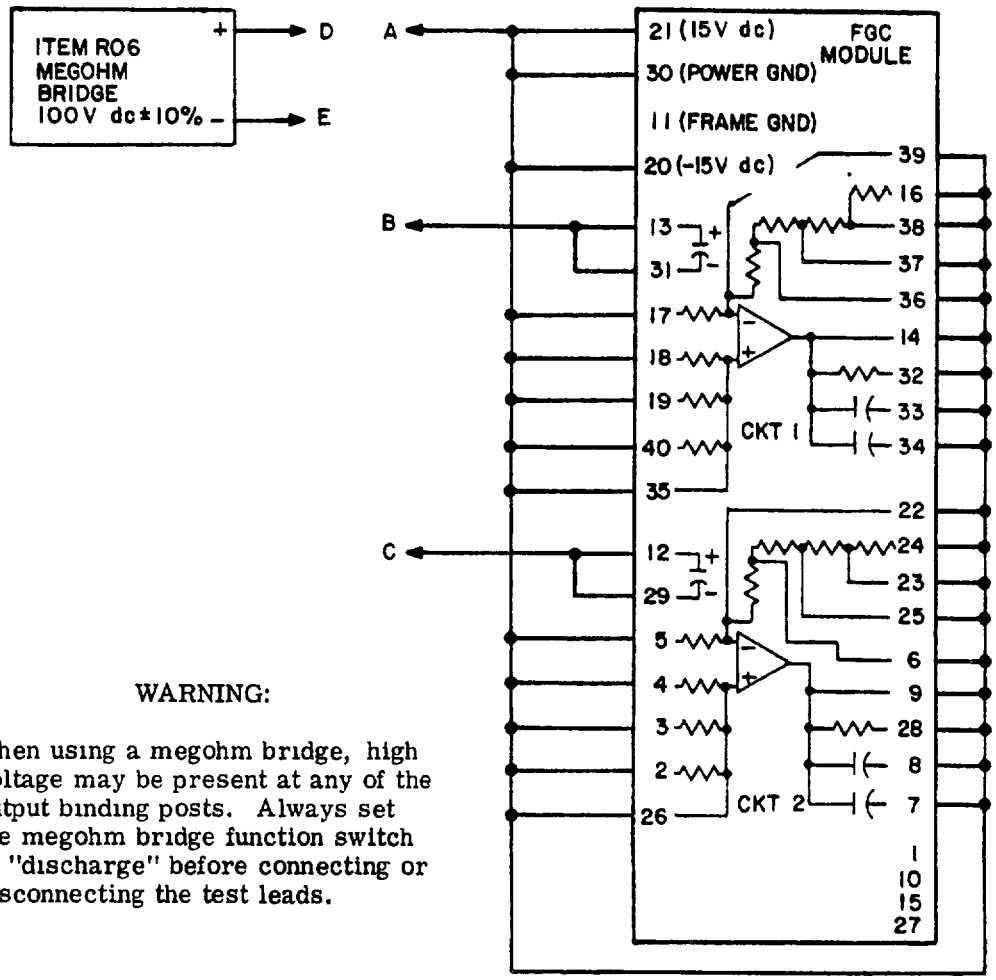
* FIGURE 7 Output short circuit current test circuit for table IV

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		D to pin	E to pin	Min	Max		
C1	Isolation, shield to module pins	A, B, C	11	10	∞	MΩ	8
C2	Isolation, circuit to circuit	B, C	A	"	"	"	"
C3	Isolation, circuit to circuit	C	B	"	"	"	"
C4	Isolation, used to unused pins	A, B, C, 11	1, 10, 27, 15	"	"	"	"
C5	Isolation, unused to unused pins	1	10, 27, 15	"	"	"	"
C6	Isolation, unused to unused pins	10	27, 15	"	"	"	"
C7	Isolation, unused to unused pins	27	15	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

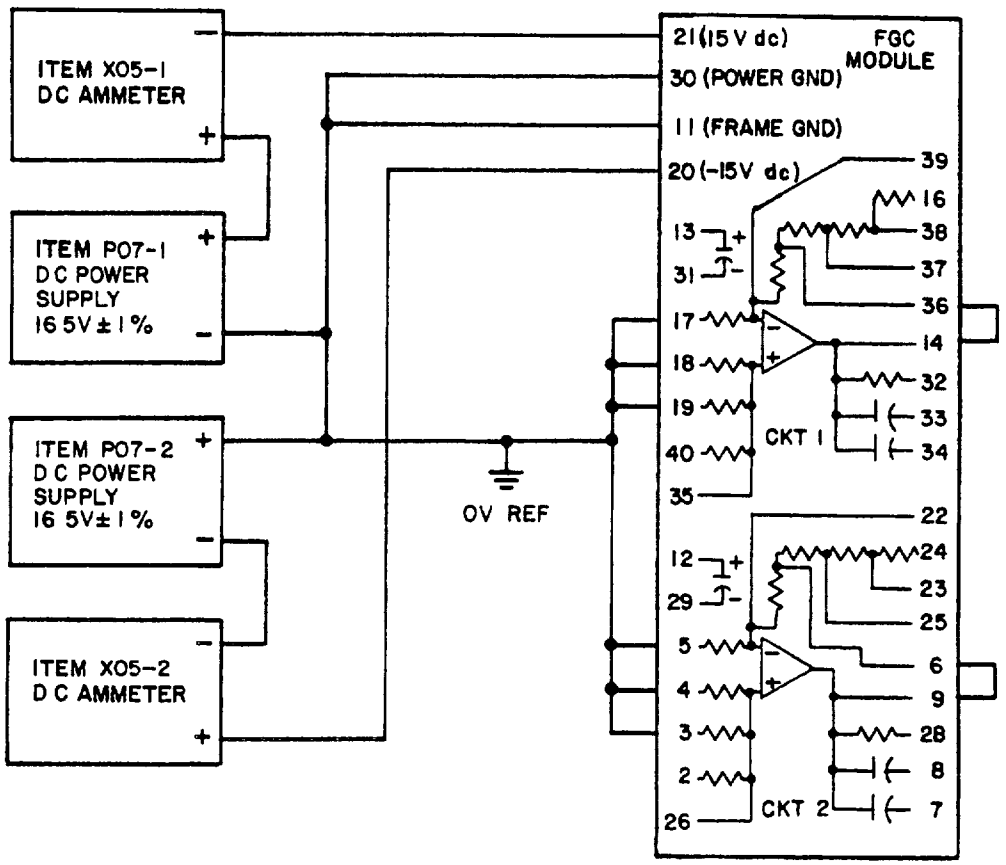
FIGURE 8. Isolation test circuit for table V.

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TABLE VI. Power supply current test.

Test condition	Parameter	Initial limits 25°C±5°C ambient		Unit	Test figure
		Min	Max		
D1	+ Power supply current	0.5	8.0	mA	9
D2	- Power supply current	0.5	8.0	mA	9

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* FIGURE 9. Power supply current test circuit for table VI.

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* TABLE VII. Input offset voltage and input offset current test.

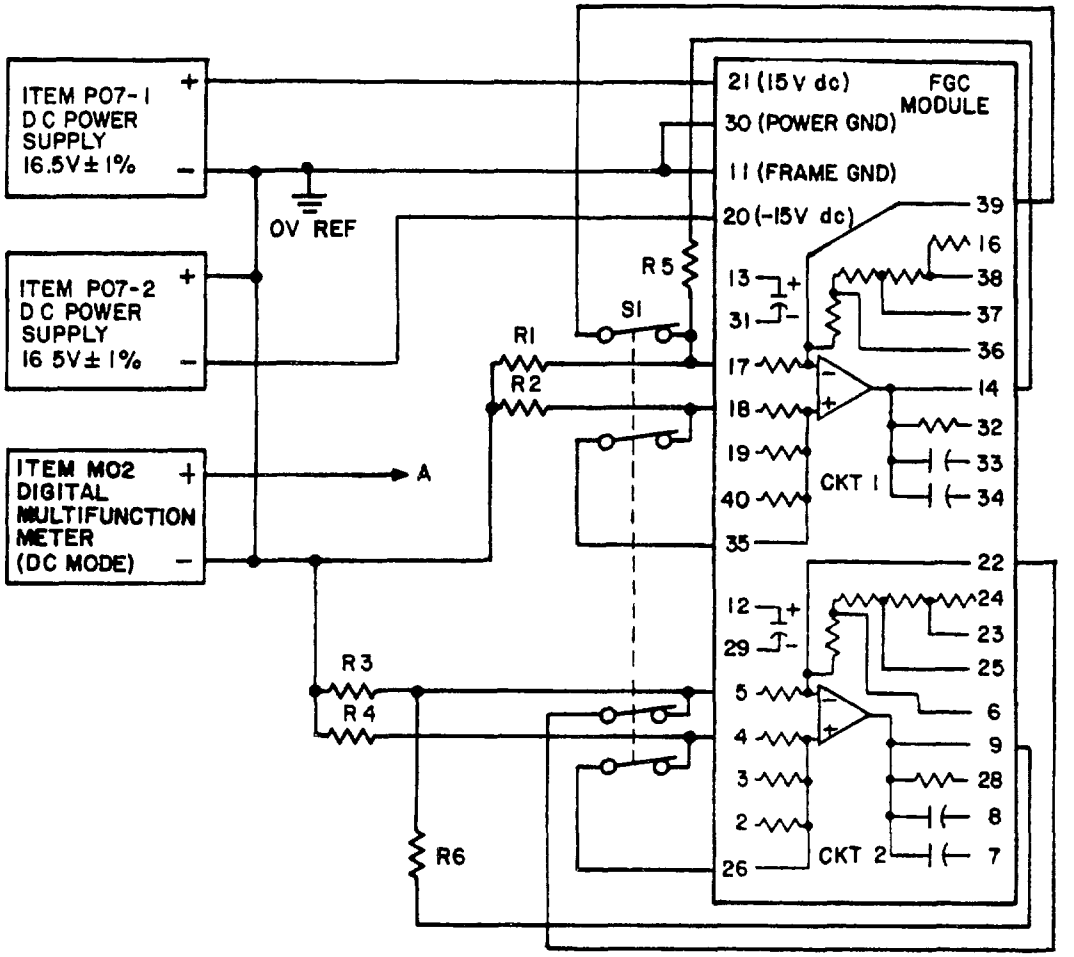
Test condition	Parameter	Connect	S1 position	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin		Min	Max		
E1	Input offset voltage (R source = 100Ω)	14	Closed	-500	500	mV	10
E2	Input offset voltage (R source = 10 kΩ)	14	Open	"	"	"	"
E3	Input offset voltage (R source = 100Ω)	9	Closed	"	"	"	"
E4	Input offset voltage (R source - 10 kΩ)	9	Open	"	"	"	"
F1	Input offset current (F1 = E1-E2)	-	-	-50	50	"	"
F2	Input offset current (F2 = E3-E4)	-	-	-50	50	"	"

1/ Test condition F is a calculation. For test condition F1, subtract the voltage measured in test condition E2 from that measured in E1. For test condition F2, subtract the voltage measured in test condition E4 from that measured in E3.

2/ $$V_{IO} = \frac{V_{OUT}}{\frac{R5+R1}{R1}} \approx \frac{V_{OUT}}{100}$$

3/ $$I_{IO} = \frac{\Delta V_{OUT}}{A_V \times R_{SERIES}} = \frac{\text{Test condition E1 - Test condition E2}}{100 \times 10\ k\Omega}.$$

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NOTES:
1. R1-R4 = 100Ω ±0.1%, 1/8 W.
2. R5 = R6 = 10 kΩ ±0.1%, 1/8 W.
3 S1 = SP4P switch.
4 For test conditions E1-E4, verify an input offset voltage of ±6 mV maximum.
5. For test conditions F1 and F2, verify an input offset current of ±50 nA maximum.

* FIGURE 10 Input offset voltage and input offset current test circuit for table VII.

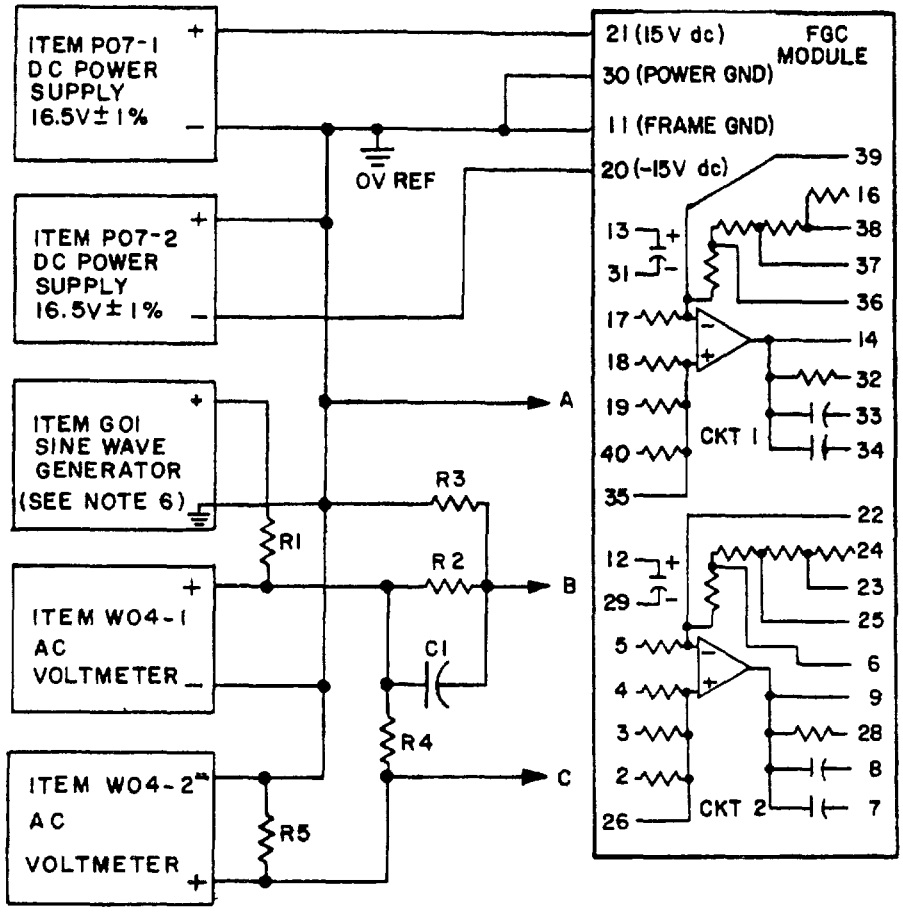
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* TABLE VIII. Large signal voltage gain test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Large signal voltage gain	35	39	14	2.4	14.0	mV	11
G2	Large signal voltage gain	26	22	9	2.4	14.0	mV	11

NOTE: Large signal voltage gain = $-100\left(\frac{7\ V}{\text{test condition G}}\right)$.

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NOTES
1. R1 = R4 = 49 9 kΩ ±1%, 1/8 W.
2 R2 = 100 kΩ ±1%, 1/8 W
3. R3 = 1 0 kΩ ±1%, 1/8 W.
4. R5 = 10 kΩ ±1%, 1/4 W
5 C1 = 470 pF ±10%, 50 V dc
6 For each test condition.
   a. Adjust item G01 frequency to 10 Hz ±5% and amplitutde to 7 V rms at point C as measured on item W04-2 with points A, B, and C connected.
   b Read item W04-1
7. Since items W04-1 and W04-2 are not used simultaneously, only one item W04 is required.

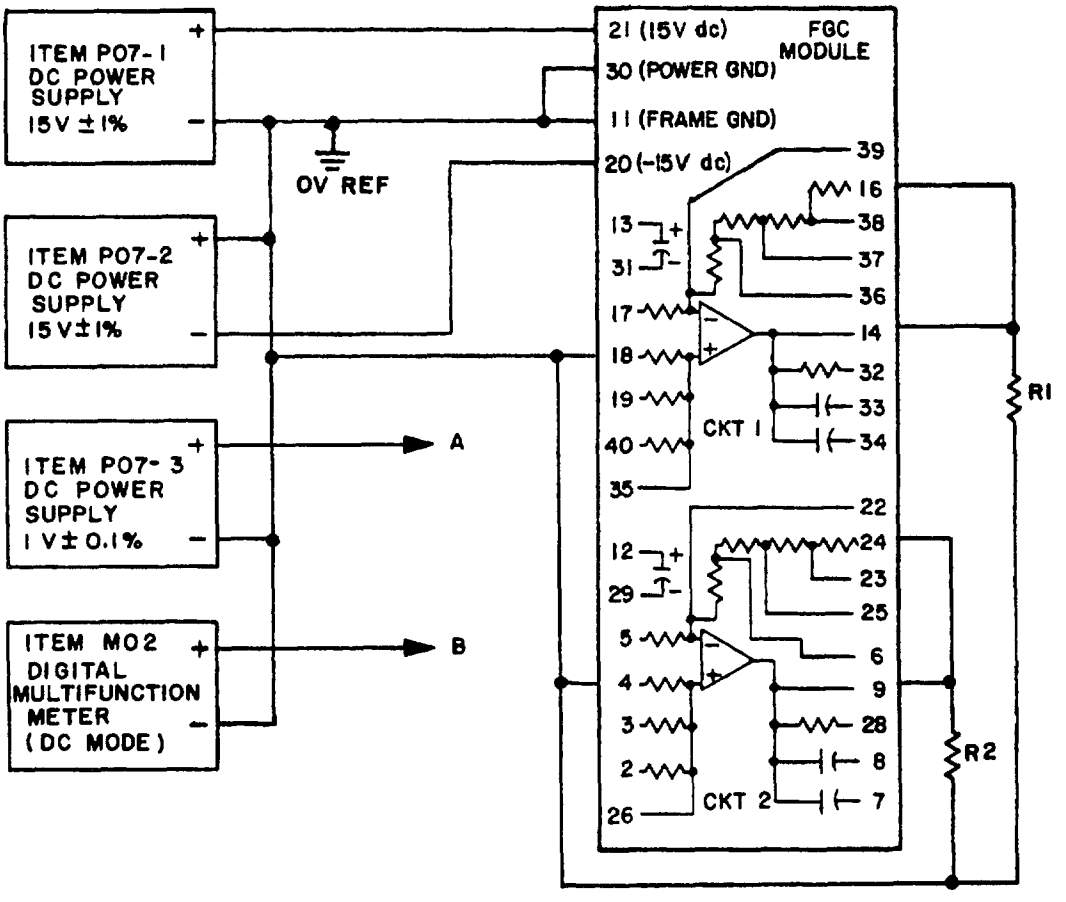
* FIGURE 11 Large signal voltage gain test circuit for table VIII

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* TABLE IX. Gain functional test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Gain functional	17	14	-10.08	-9.92	V	12
H2	Gain functional	5	9	-10.08	-9.92	V	12

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NOTE. R1 = R2 = 10 kΩ ±1%, 1/8 W.

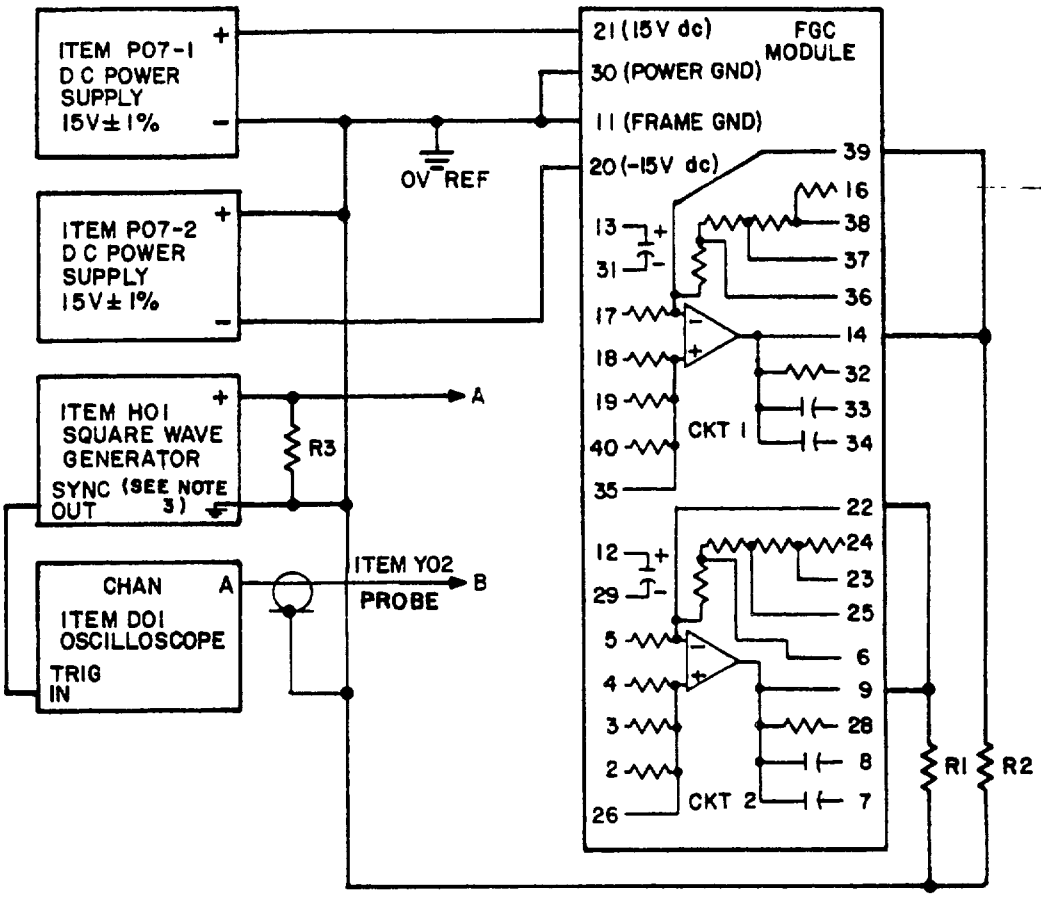
FIGURE 12. Gain functional test circuit for table IX.

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TABLE X. Slew rate test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Slew rate (neg to pos)	35	14	0.1	8.3	µs	13, 14
J2	Slew rate (pos to neg)	35	14	"	"	"	"
J3	Slew rate (neg to pos)	26	9	"	"	"	"
J4	Slew rate (pos to neg)	26	9	"	"	"	"

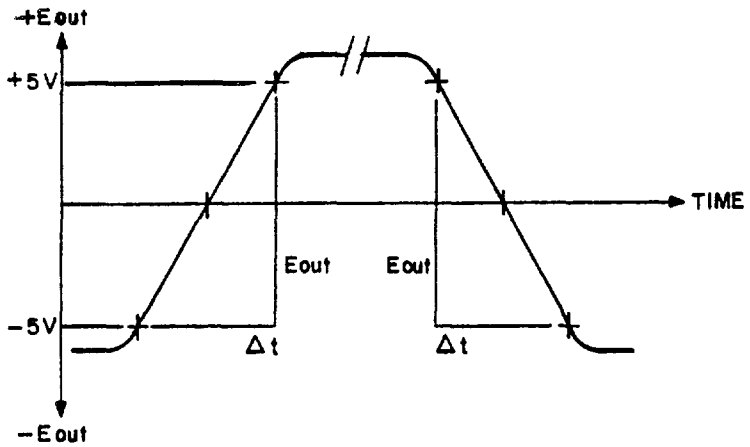
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NOTES.
1. R1 = R2 = 2 kΩ ±1%, 1/4 W.
2. R3 = 600Ω ±5%, 1 W.
3. Item H01 setup:
   Frequency = 20 kHz.
   Output at A = 20 V peak to peak.
4. This test verifies a slew rate of 1.2 V/µs minimum.

FIGURE 13. Slew rate test circuit for table X.

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$$\text{SLEW RATE} = \frac{\Delta E_{out}}{\Delta t} = \frac{10}{\Delta t}$$

$$\Delta t = \frac{10V}{1.2V/\mu s} = 8.3\mu s \text{ MAXIMUM}$$

FIGURE 14. Slew rate test circuit waveforms for table X.

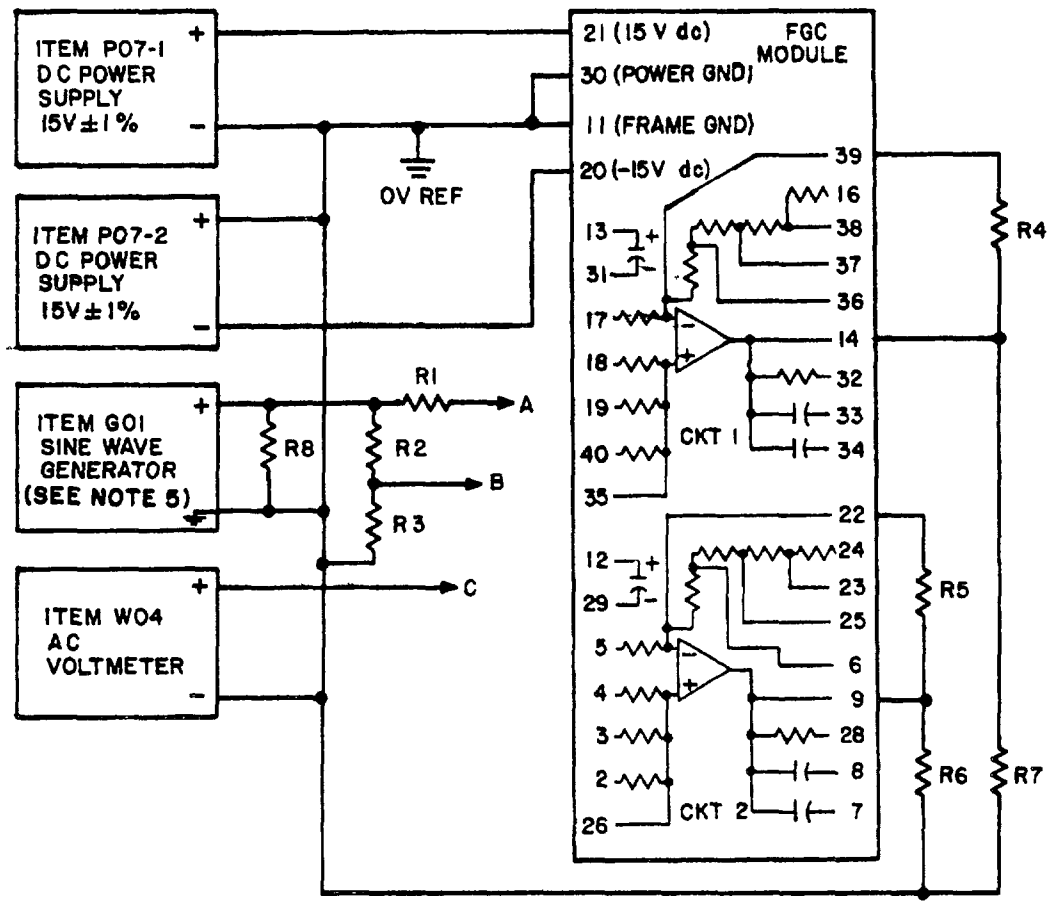
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TABLE XI. Common mode rejection ratio test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Common mode rejection ratio	39	35	14	0	100	mV	15
K2	Common mode rejection ratio	22	26	9	0	100	mV	15

NOTE: $CMRR = 20 \log\left(\frac{V_{OUT(avg)}}{V_{IN}}\right) = 20 \log\left(\frac{\text{Item W04 reading in volts/1,000}}{1 \text{ V rms}}\right)$.

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NOTES
1. R1 = R2 = 100Ω ±0.1%, 1/8 W, |R1-R2| ≤ 1Ω.
2. R3 = R4 = R5 = 100 kΩ ±0.01%, 1/8 W, |R3-R4| ≤ 10Ω, |R3-R5| ≤ 10Ω
3. R6 = R7 = 2 kΩ ±1%, 1/4 W.
4. R8 = 49.9Ω ±5%, 1 W.
5. Item G01 setup.
   Frequency = 1 kHz ±5%.
   Output at A = 1 V rms ±1%.
6. This test verifies a common mode rejection ratio of -80 dB maximum.

FIGURE 15. Common mode rejection ratio test circuit for table XI.

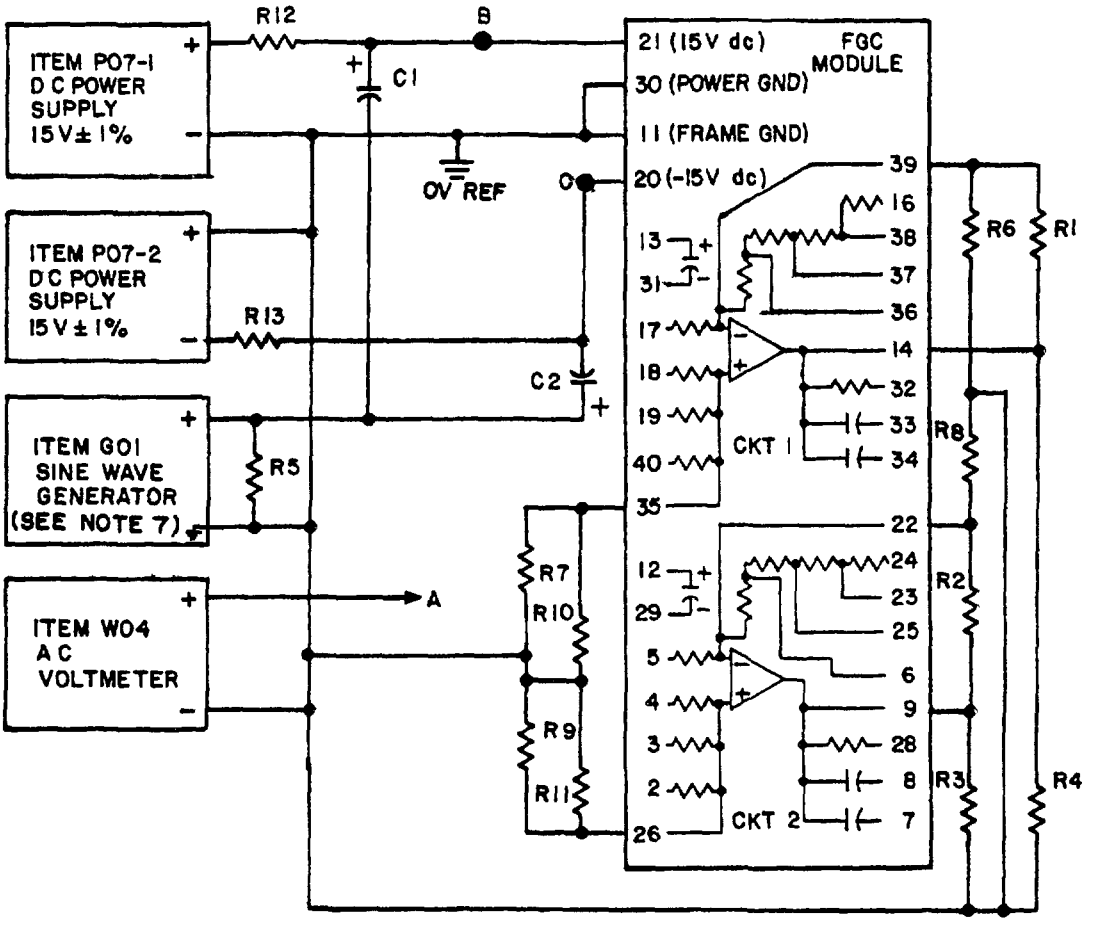
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TABLE XII. Supply voltage rejection ratio test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Supply voltage rejection ratio	14	0	500	mV	16
L2	Supply voltage rejection ratio	9	0	500	mV	16

NOTE: $PSRR = 20 \log\left(\frac{V_{OUT(avg)}}{V_{IN}}\right) = 20 \log\left(\frac{\text{Item W04 reading in volts/1,000}}{1 \text{ V rms}}\right)$.

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NOTES.
1. R1 = R2 = R10 = R11 = 100 kΩ ±0 1%, 1/8 W
2. R3 = R4 = 2 kΩ ±1%, 1/4 W
3. R5 = 49.9Ω ±5%, 1 W.
4. R6 = R7 = R8 = R9 = 100Ω ±0.1%, 1/8 W
5. R12 = R13 = 100Ω ±1%, 1/8 W
6. C1 = C2 = 20 µF ±20%, 20 V dc, electrolytic.
7. Item G01 setup:
   Frequency = 1 kHz ±5%.
   Output = 1 V rms ±5% at points B and C.
8. This test verifies a supply voltage rejection ratio of -66 dB maximum.

FIGURE 16. Supply voltage rejection ratio test circuit for table XII.

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TABLE XIII. Small signal bandwidth and gain peaking test.

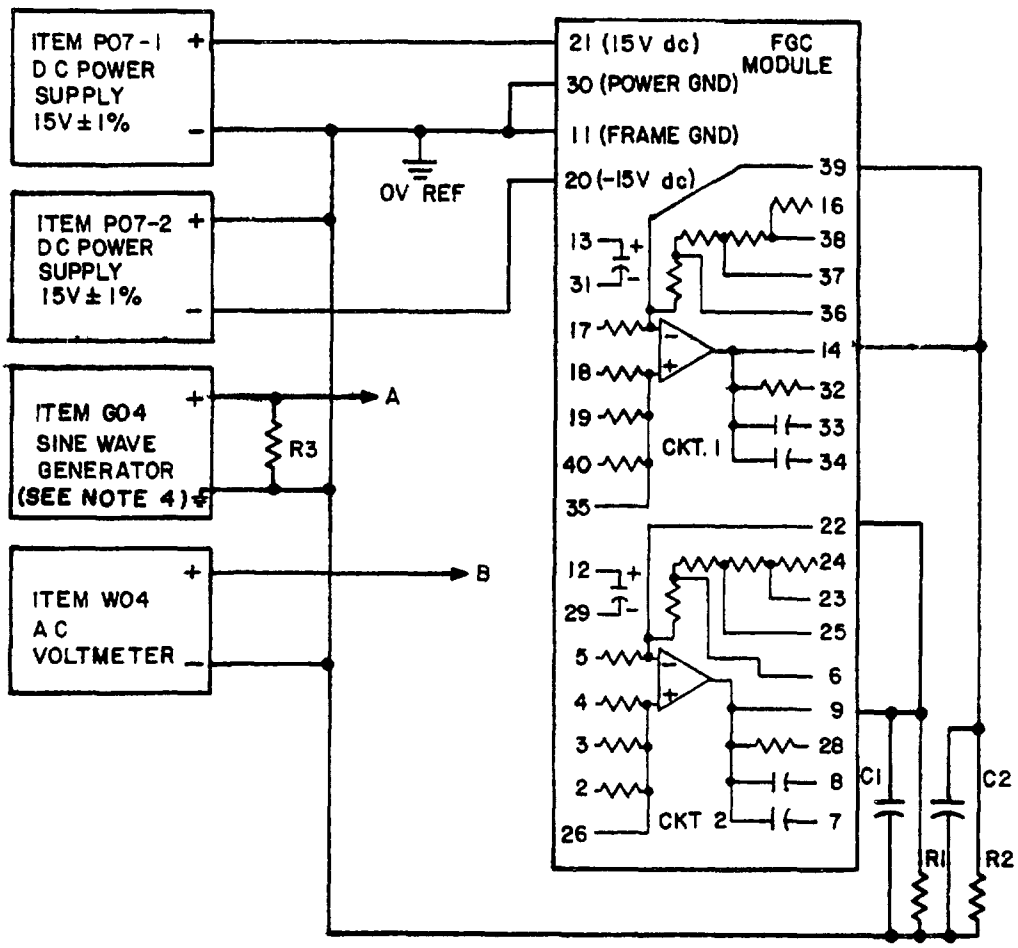
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Small signal bandwidth	35	14	3	10	MHz	17
M2	Small signal bandwidth	26	9	3	10	MHz	"
M3	Gain peaking	35	14	20	50	mV	"
M4	Gain peaking	26	9	20	50	mV	"

NOTES:

1. Gain peaking ratio (table I) = $\frac{V_{OUT}}{V_{IN}} = \frac{V_{OUT}}{20 \text{ mV rms}}$.

2. This test verifies a minimum phase margin $\theta m \approx 20^{\circ}$.

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NOTES
1 R1 = R2 = 2 kΩ ±1%, 1/4 W.
2. R3 = 49.9Ω ±5%, 1 W.
3 C1 = C2 = 100 pF ±20%.
4 For each test condition:
a. Adjust item G04 to yield 20 mV rms and 1 kHz at point B.
b. For test conditions M1 and M2, maintain input voltage amplitude within ±1%, continuously increase frequency until output voltage rolls off to 14.14 mV rms (-3 dB point).
c Item G04 frequency at this -3 dB point shall conform to table XIII requirements
d For test conditions M3 and M4, maintain input voltage within ±1%; continuously increase frequency to 10 MHz. Maximum output voltage shall conform to table XIII requirements

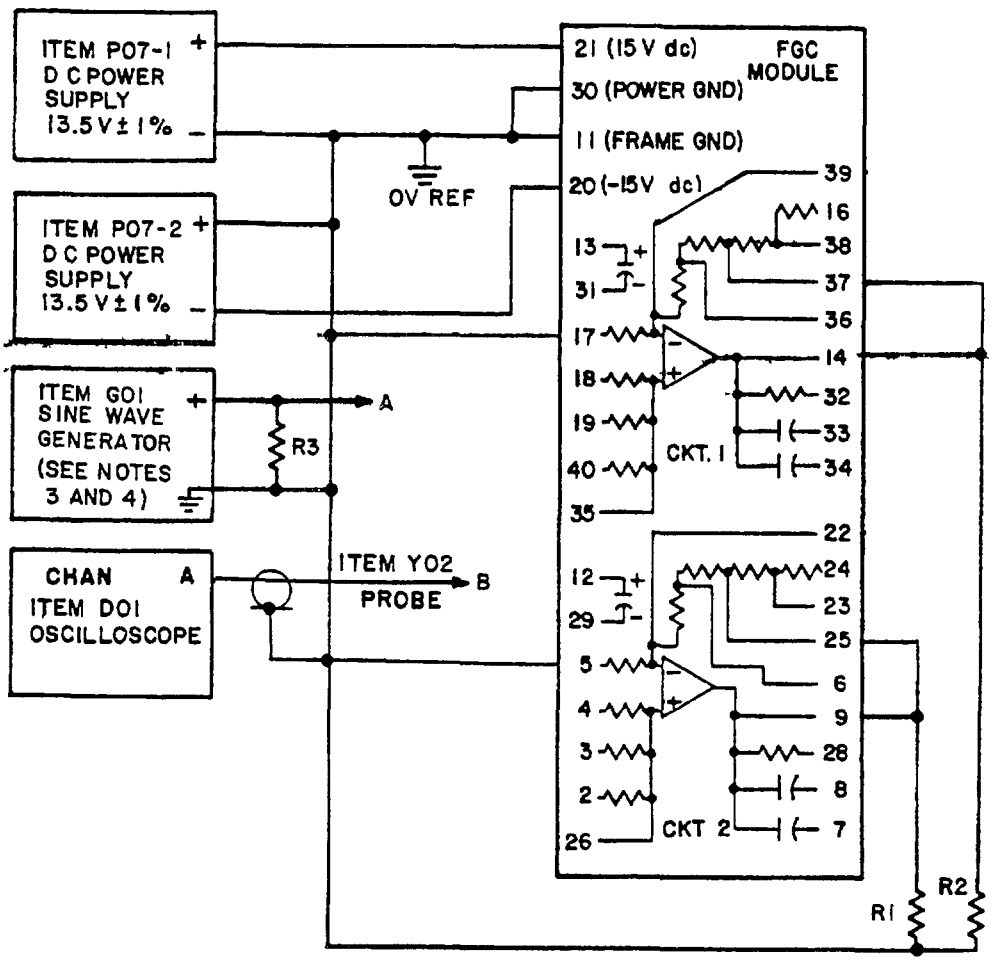
FIGURE 17. Small signal bandwidth and gain peaking test circuit for table XIII.

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TABLE XIV. Output voltage swing test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Output voltage swing (+)	35	14	8.5	13.5	Vpk	18
N2	Output voltage swing (+)	26	9	8.5	13.5	"	"
N3	Output voltage swing (-)	35	14	-13.5	-8.5	"	"
N4	Output voltage swing (-)	26	9	-13.5	-8.5	"	"

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NOTES.
1. R1 = R2 = 569Ω ±1%, 1/4 W.
2. R3 = 49.9Ω ±5%, 1 W.
3. Item G01 frequency = 20 kHz.
4. For each test condition:
   a. Continuously increase item G01 amplitude until clipping occurs at point B.
   b. For test conditions N1 and N2, this clip point shall be measured on the positive side of the voltage swing.
   c. For test conditions N3 and N4, this clip point shall be measured on the negative side of the voltage swing.
5. This test also verifies output drive current capability of 15 mA minimum.

FIGURE 18. Output voltage swing test circuit for table XIV.

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TABLE XV. Noise test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
P1	Noise	14	0	1	mV rms	19
P2	Noise	0	0	1	mV rms	19

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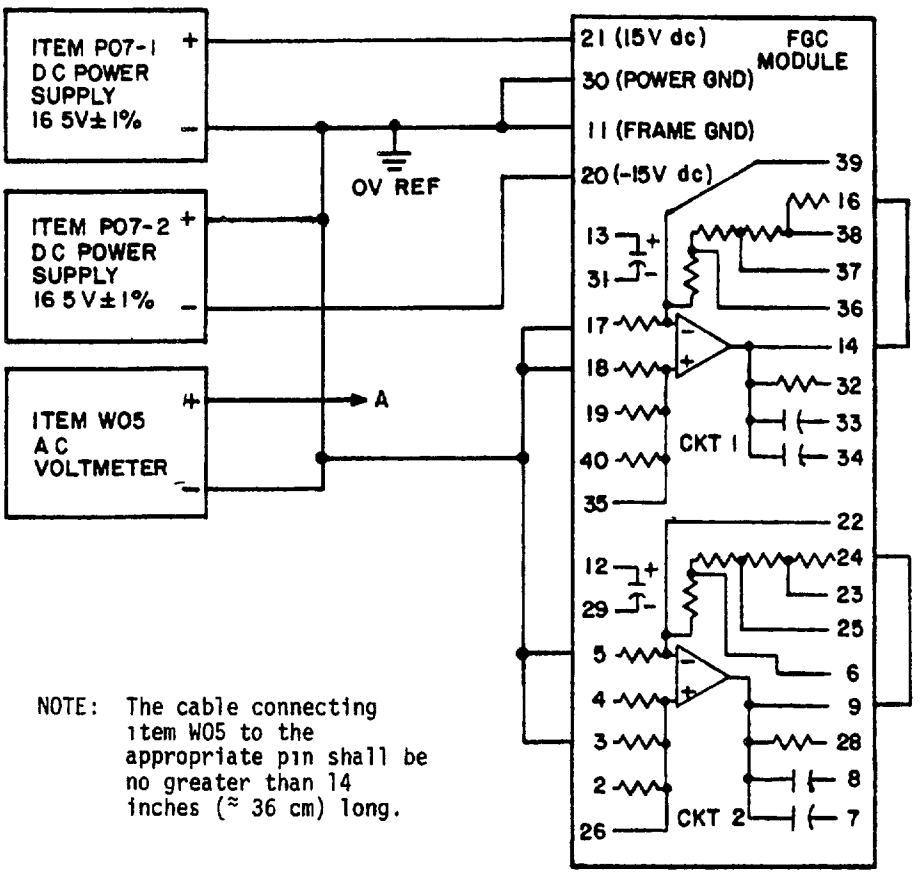


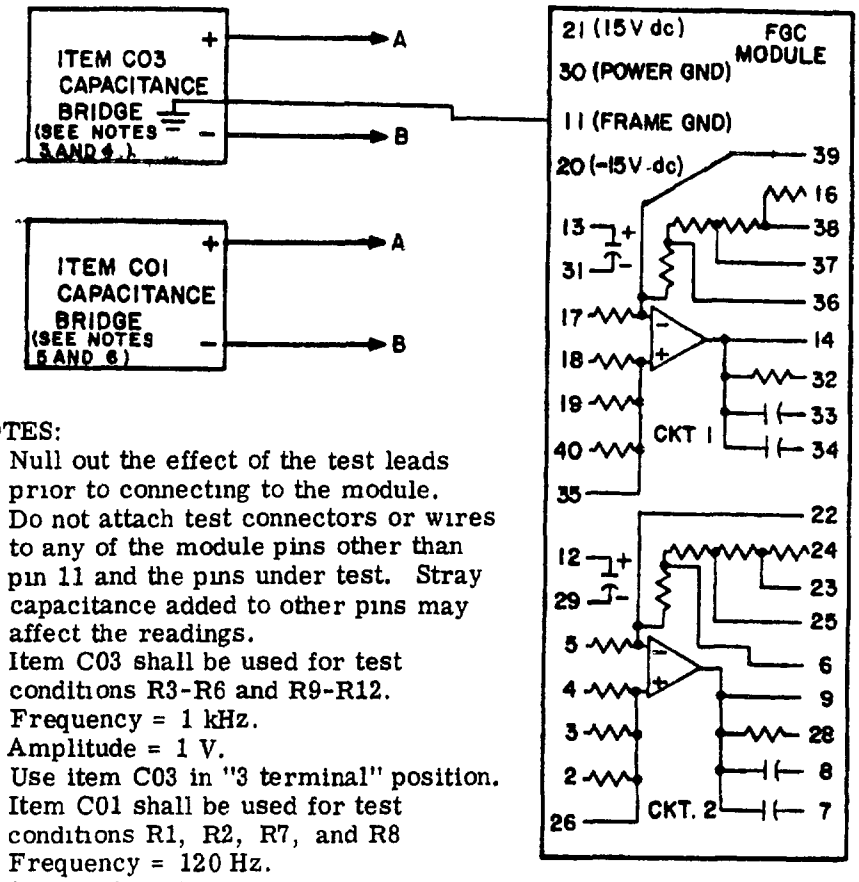
FIGURE 19. Noise test circuit for table XV.

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TABLE XVI. Capacitance, dissipation factor, and power supply capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Capacitance (C1)	13	31	1.9	2.1	µF	20
R2	Capacitance (C2)	12	29	1.9	2.1	µF	"
R3	Capacitance (C3)	14	33	76	84	pF	"
R4	Capacitance (C5)	9	8	76	84	"	"
R5	Capacitance (C4)	14	34	152	168	"	"
R6	Capacitance (C6)	9	7	152	168	"	"
R7	Dissipation factor (C1)	13	31	0	5	%	"
R8	Dissipation factor (C2)	12	29	"	5	"	"
R9	Dissipation factor (C3)	14	33	"	3	"	"
R10	Dissipation factor (C5)	9	8	"	"	"	"
R11	Dissipation factor (C4)	14	34	"	"	"	"
R12	Dissipation factor (C6)	9	7	"	"	"	"
R13	Power supply capacitance (+)	21	30	0.075	0.125	µF	"
R14	Power supply capacitance (-)	30	20	0.075	0.125	µF	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connectors or wires to any of the module pins other than pin 11 and the pins under test. Stray capacitance added to other pins may affect the readings.
3. Item C03 shall be used for test conditions R3-R6 and R9-R12.
   Frequency = 1 kHz.
   Amplitude = 1 V.
4. Use item C03 in "3 terminal" position.
5. Item C01 shall be used for test conditions R1, R2, R7, and R8
   Frequency = 120 Hz.
   Amplitude = 1 V.
6. Item C01 shall be used for test conditions R13 and R14.
   Frequency = 120 Hz.
   Amplitude = 0.3 Vpk maximum.

FIGURE 20. Capacitance, dissipation factor, and power supply capacitance test circuit for table XVI

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TABLE XVII. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	Resistance (R1)	39	17	9.98	10.02	kΩ	21
S2	Resistance (R1+R5)	36	"	19.96	20.04	"	"
S3	Resistance (R1+R5+R6)	37	"	39.92	40.08	"	"
S4	Resistance (R1+R5+R6+R7)	38	"	59.88	60.12	"	"
S5	Resistance (R1+R5+R7+R8)	16	"	109.68	110.22	"	"
S6	Resistance (R2)	18	35	9.98	10.02	"	"
S7	Resistance (R3)	19	"	9.98	10.02	"	"
S8	Resistance (R4)	40	"	39.92	40.08	"	"
S9	Resistance (R3+R4)	40	19	49.90	50.10	"	"
S10	Resistance (R2+R4)	40	18	49.90	50.10	"	"
S11	Resistance (R2+R3)	19	18	19.96	20.04	"	"
S12	Resistance (R9)	32	14	95.00	105.00	Ω	"
S13	Resistance (R10)	22	5	9.98	10.02	kΩ	"
S14	Resistance (R10+R14)	6	"	19.96	20.04	"	"
S15	Resistance (R10+R14+R15)	25	"	39.92	40.08	"	"
S16	Resistance (R10+R14+R15+R16)	23	"	59.88	60.12	"	"
S17	Resistance (R10+R14+R15+R16+R17)	24	"	109.68	110.22	"	"
S18	Resistance (R11)	4	26	9.98	10.02	"	"

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TABLE XVII. Resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S19	Resistance (R12)	3	26	9.98	10.02	kΩ	21
S20	Resistance (R13)	2	26	39.92	40.08	"	"
S21	Resistance (R12+R13)	"	3	49.90	50.10	"	"
S22	Resistance (R11+R13)	"	4	49.90	50.10	"	"
S23	Resistance (R11+R12)	3	4	19.96	20.04	"	"
S24	Resistance (R18)	28	9	95.00	105.00	Ω	"

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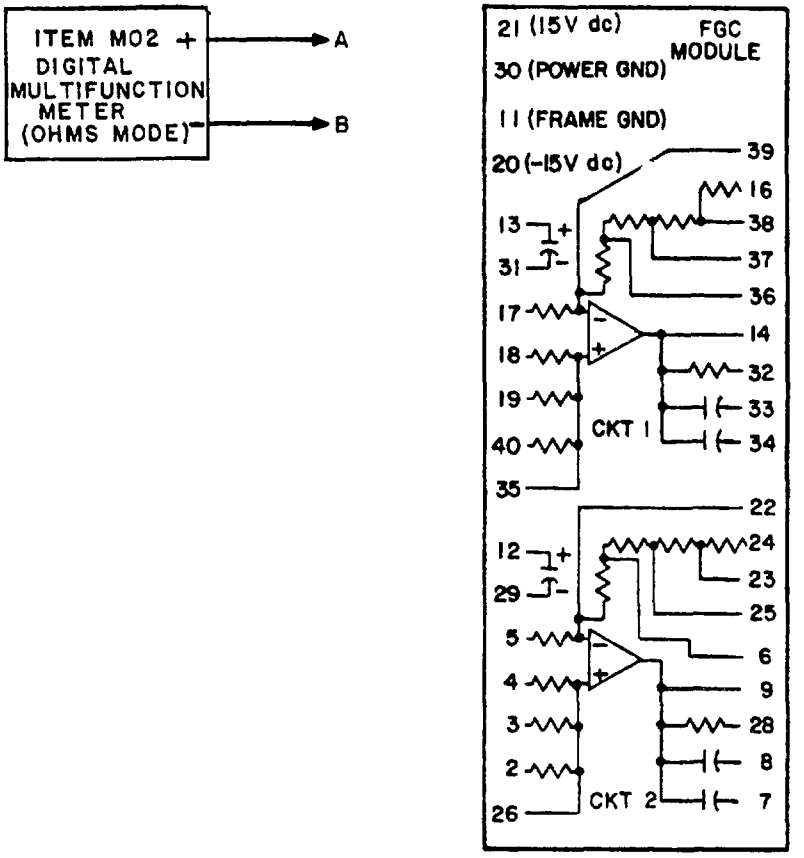


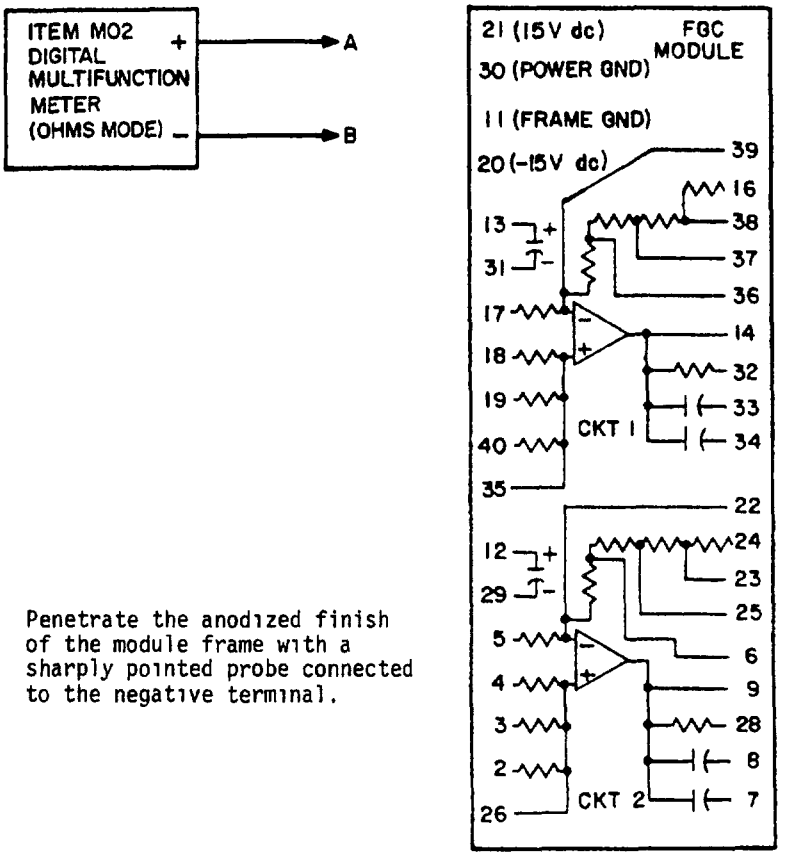
FIGURE 21. Resistance test circuit for table XVII.

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TABLE XVIII. Frame continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Frame continuity	11	Frame	0	1	Ω	22

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NOTE. Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 22. Frame continuity test circuit for table XVIII.

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* TABLE XIX. Input bias current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U1	Input bias current	35	-2.4	2.4	mV	23
U2	Input bias current	26	-2.4	2.4	mV	23

NOTES:

1. Average input bias current (table I) = $\frac{|\text{test condition U}|}{10\ \text{k}\Omega \times 2}$.
2. This test in conjunction with test condition F (offset current test) verifies an input bias current ≤ 145 nA per input where, $I_{IB(max)} = |I_{IB(avg)}| + \frac{|I_{IO}|}{2}$.

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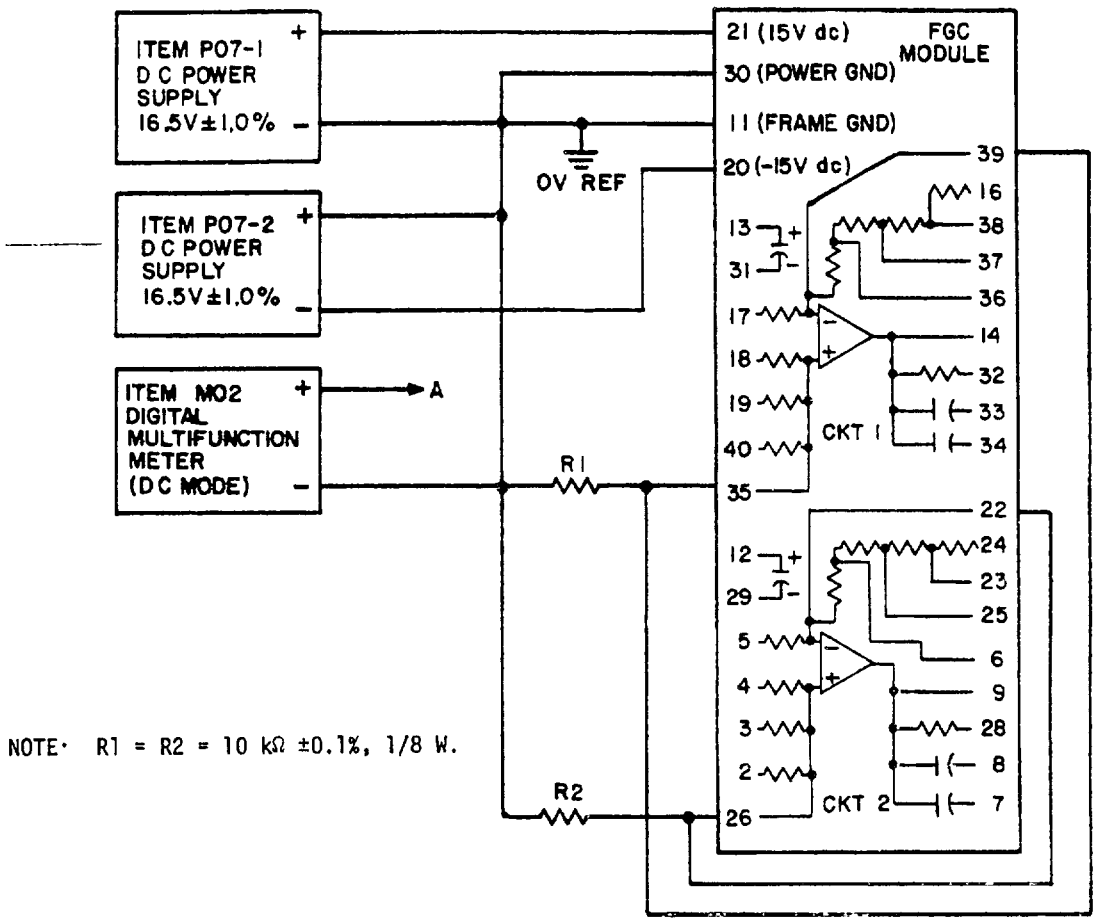


FIGURE 23. Input bias current test circuit for table XIX.

Source: https://assist.dla.mil -- Downloaded: 20260-20T16:28Z
Check the source to verify that this is the current version before use.

Preparing activity:
Navy - EC

Agent:
DLA- ES

(Project 5963-N676)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:28Z
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